Electric Vehicle Market Performance & Insights

Data Cleaning and Preparation

Repository: https://github.com/manasgite54-cmd/EV-project

Purpose and Context

The reported pipeline writes the cleaned CSV and JSON report. This document records the reproducibility expectations; exact operations and counts must be taken from the implemented code and report rather than assumed.

Detailed Project-Specific Content

- Input inventory: record source path, format, encoding, delimiter, schema and original row count.
- Schema validation: check required columns and stop or report clearly if the schema changes.
- Type conversion: apply explicit conversion rules to numeric/year fields; report failed conversions.
- Missingness: calculate null/blank rates by field and use field-specific rules; avoid blanket row deletion without measuring impact.
- Duplicate handling: define duplicate keys and count candidates before removal; repeated listings may be legitimate depending on row grain.
- Category normalization: trim whitespace and standardize only labels known to be equivalent.
- Range/outlier checks: flag implausible values for review; do not delete rare but valid vehicles automatically.
- Export processed data to data/processed/ev_market_clean.csv and cleaning statistics to data/processed/cleaning_report.json.
- Reconcile input/output row counts and verify report totals against the CSV.
- Keep raw data unchanged, use deterministic steps where possible, document dependencies and provide the command needed to rerun.
- Reported output count is 299,999. Rule-level counts, rows removed, null rates and outlier counts must be transcribed from the actual JSON report.

Project References and Artifact Locations

Project: Electric Vehicle Market Performance & Insights Repository: https://github.com/manasgite54-cmd/EV-project Processed dataset: data/processed/ev_market_clean.csv Reported processed records: 299,999 Cleaning report: data/processed/cleaning_report.json Tableau Dashboard: https://public.tableau.com/views/Book1_17931030431560/EVMarketOverview?:showVizHome=no&:embed=yes Tableau Story: https://public.tableau.com/views/Book1_17931030431560/Story1?:showVizHome=no&:embed=yes Accuracy note: the record count and artifact paths above are reported by the project owner. Exact schema, source provenance, units, calculated statistics, test results, and deployment status must be checked against the repository and actual artifacts. No unsupported results, approvals, or signatures are claimed.

Status: Detailed project-specific document. Any [TO BE PROVIDED], [TO BE VERIFIED] or [TO BE CALCULATED] item requires confirmation. No supervisor approval, signature, unverified test pass or successful deployment is implied.